

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Senior Quality Engineering professional with over **7+ years** of experience defining quality strategy and building scalable automation for modern SaaS, healthcare, enterprise, API-driven, and workflow automation platforms.
- Strong hands-on expertise with **Playwright, TypeScript, JavaScript, Python, C#, Selenium WebDriver, REST APIs, SQL, and YAML-based automation**, developing reliable web, API, backend, integration, and end-to-end test suites.
- Experienced designing and maintaining reusable test frameworks for **microservice-based architectures**, integrating automated validation throughout CI/CD pipelines to reduce manual regression effort and provide fast, actionable release feedback.
- Strong experience testing **AI/LLM-enabled applications, healthcare software, enterprise SaaS products, and workflow automation platforms**, with focus on reliability, data integrity, compliance, integration behavior, and production readiness.
- Hands-on experience with **Git, GitHub, GitHub Actions, Docker, Kubernetes, AWS, PostgreSQL, React, Node.js, and cloud-native architectures**, supporting automated testing across distributed application environments.
- Experienced defining test plans, acceptance criteria, release-readiness standards, production quality metrics, and observability signals while performing exploratory, functional, regression, integration, performance, and end-to-end validation.
- Quality-focused technical leader experienced in code reviews, debugging, root-cause analysis, mentoring engineers, investigating customer-reported issues, and partnering across Engineering, Product, and Design to build a culture of shared quality ownership.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Application Stack:** TypeScript, JavaScript, Python, C#, Java, SQL, YAML, React, Node.js, REST APIs, JSON, Object-Oriented Programming, Microservices.
- **Automation Frameworks:** Playwright, Cypress, Selenium WebDriver, Pytest, NUnit, XUnit, JUnit, TestNG, Cucumber BDD, RESTAssured, Page Object Model, Reusable Test Architecture.
- **Testing Practices:** Exploratory Testing, Functional Testing, Regression Testing, Integration Testing, End-to-End Testing, API Testing, Backend Testing, Performance Testing, Load Testing, Security Testing.
- **Data & Integration:** PostgreSQL, SQL Server, MySQL, SQL Queries, Data Validation, Data Reconciliation, REST API Testing, Microservice Validation, Workflow Automation Testing, Backend Service Testing.
- **DevOps & Cloud:** Git, GitHub, GitHub Actions, Azure DevOps, Jenkins, CI/CD Pipelines, YAML Pipelines, Docker, Kubernetes, AWS, Automated Quality Gates, Deployment Validation.
- **AI & Observability:** AI/LLM Testing, Generative AI Integrations, Prompt/Response Validation, AI-Assisted Automation, Datadog, Grafana, Logging, Metrics, Monitoring, Production Quality Analysis.
- **Quality Engineering Leadership:** Quality Strategy, Test Plans, Acceptance Criteria, Release Readiness, Code Reviews, Root Cause Analysis, Quality Metrics, Risk Analysis, Mentoring, Agile/Scrum, Continuous Improvement.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

San Francisco, CA, USA

- Own quality strategy across enterprise SaaS features by defining **test plans, acceptance criteria, automation scope, risk-based coverage, and release-readiness expectations** in partnership with Engineering, Product Owners, and business stakeholders.
- Design and maintain scalable automation frameworks using **Playwright, TypeScript/C#, Python, REST APIs, SQL, and YAML-based configuration**, covering web interfaces, backend services, integrations, and end-to-end business workflows.
- Build API and microservice validation using Postman, REST/JSON, SQL Server/PostgreSQL-style queries, and reconciliation checks to verify contracts, business rules, data integrity, and cross-service consistency across integrated applications.
- Integrate automated testing with **Azure DevOps/TeamCity, Git, Docker, and CI/CD pipelines**, enabling parallel execution, automated quality gates, repeatable regression coverage, and faster deployment feedback with reduced manual regression effort.
- Validate AI/LLM-enabled capabilities through structured input/output checks, business-rule validation, regression scenarios, and edge-case analysis while using AI-assisted tools to accelerate test generation, debugging, and automation maintenance.
- Monitor production-quality signals through logs, metrics, API telemetry, defect trends, and customer-reported issues, performing root-cause analysis and partnering with developers to implement durable fixes and stronger regression protection.
- Champion engineering excellence through automation code reviews, reusable framework standards, mentoring, quality-metric reporting, and continuous process improvements that increase test reliability, maintainability, and shared ownership of quality.

Environments: Playwright, TypeScript, C#, Python, REST APIs, Postman, SQL Server/PostgreSQL, YAML, Azure DevOps, TeamCity, Git, Docker, CI/CD, AI/LLM Testing, API/Microservice Testing, Quality Metrics, Production Monitoring, Root Cause Analysis, Agile/Scrum.

SDET / QA Automation Engineer

Dec 2023 – Sept 2024

Client: CVS

San Francisco, CA, USA

- Developed scalable automated test coverage across healthcare web applications, APIs, backend services, and integrated workflows using **TypeScript, Playwright, Selenium WebDriver, Python/Java, RESTAssured, and SQL**.
- Designed functional, exploratory, integration, regression, and end-to-end strategies for regulated healthcare workflows while partnering with developers and Product stakeholders to identify quality risks during feature design and implementation.
- Validated REST APIs and microservice interactions using **Postman, RESTAssured, JSON, SQL, and backend reconciliation**, ensuring accurate service contracts, data transformations, business processing, and integration behavior.
- Integrated automated regression with **Jenkins, GitHub/GitLab, Git, YAML-based CI/CD pipelines, and containerized environments**, supporting continuous testing, fast feedback, automated quality gates, and reliable release decisions.
- Tested AI-assisted and data-driven application capabilities using controlled scenarios, structured datasets, API validation, and expected business outcomes while applying AI tools to improve test creation and failure investigation.
- Investigated flaky tests, customer-impacting issues, and integration failures using Splunk, application logs, API traces, SQL, and root-cause analysis, strengthening automation coverage around recurring defects and high-risk workflows.
- Worked with Engineering, Product, DevOps, and QA teams to review automation code, communicate quality metrics and release risks, mentor peers, and continuously improve testing practices within fast-paced Agile delivery.

Environments: TypeScript, Playwright, Selenium WebDriver, Python, Java, RESTAssured, Postman, JSON, SQL, Jenkins, GitHub/GitLab, Git, YAML, CI/CD, Docker, Healthcare Applications, AI-Assisted Testing, Splunk, Quality Metrics, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and maintained reusable automation solutions using **Python, Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL** across enterprise SaaS applications and distributed backend services.
- Built automated functional, integration, regression, API, and end-to-end coverage while partnering with engineering teams to identify testability concerns, missing scenarios, and integration risks early in feature development.
- Performed backend and microservice validation through Postman, REST/SOAP services, JSON/XML, SQL, and database reconciliation, verifying data transformations, business logic, cross-service communication, and workflow integrity.
- Integrated regression automation with **Jenkins, Git, Maven, Docker, and CI/CD pipelines**, enabling repeatable test execution, automated build validation, and earlier identification of quality regressions throughout deployment workflows.
- Developed reusable test-data utilities, framework helpers, service abstractions, and YAML/configuration-driven execution patterns that improved scalability, maintainability, and reuse across multiple enterprise application modules.
- Investigated application and automation failures using logs, SQL results, API responses, and debugging techniques while performing root-cause analysis and implementing improvements that increased regression stability and coverage.
- Participated in code reviews, Agile ceremonies, quality discussions, and process-improvement initiatives while mentoring peers on test design, framework maintainability, API validation, and reliable automation practices.

Environments: Python, Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, Postman, SQL, Jenkins, Git, Maven, Docker, YAML, CI/CD, Enterprise SaaS, Microservices, Framework Development, RCA, Agile/Scrum.

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed and executed automated and manual testing across enterprise web, API, backend, and integration workflows using **Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL**.
- Created reusable functional, regression, integration, and end-to-end automation suites covering business-critical application flows while reducing repetitive manual validation and increasing repeatable regression coverage.
- Performed API and backend testing using **Postman, REST/SOAP interfaces, JSON/XML, and SQL**, validating request-response contracts, business processing, data integrity, and consistency across connected systems.
- Integrated automated suites with **Jenkins, Git, and CI/CD pipelines**, supporting automated build checks, scheduled regression execution, repeatable release validation, and faster feedback throughout Agile software delivery.
- Maintained reusable framework utilities, test data, shared page components, and configuration-driven execution patterns while identifying opportunities to simplify maintenance and improve test-suite reliability.
- Investigated failed and intermittent tests through logs, screenshots, API responses, SQL records, and debugging techniques while collaborating with developers to identify root causes and verify permanent fixes.
- Worked with developers, analysts, and QA engineers during requirement reviews, code/test reviews, defect triage, and release activities while contributing to quality standards, knowledge sharing, and continuous testing improvements.

Environments: Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, REST/SOAP APIs, JSON/XML, SQL, Jenkins, Git, CI/CD, Enterprise Applications, API/Integration Testing, Regression Automation, Root Cause Analysis, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.